

Jake Golden

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EDUCATION

Duke University – Pratt School of Engineering

Bachelor of Engineering in Electrical and Computer Engineering, Biomedical Engineering
Cumulative GPA: 4.00/4.00; Dean's List with Distinction (2024, 2025, 2026).

Expected May 2027

Durham, NC

EXPERIENCE | [Click on position titles for more information](#)

Analog Design Intern

National Instruments (Emerson)

May 2026 – Aug 2026

Austin, TX

- Designed schematic and board layout (Siemens Xpedition) for precision LCR/SMU meter characterization PCB, implemented low-leakage, guarded switching topology, led schematic/layout reviews with cross-functional partners.
- Authored 15+ V&V test protocols for LCR/SMU meter performance evaluation, automated instrumentation control (oscilloscopes, DMMs, SMUs) and data acquisition/processing using LabVIEW, compiled results in formal report.

Product Development Intern

Stryker

May 2025 – Aug 2025

Mahwah, NJ

- Developed 9 prototypes for novel implant; created design inputs to address unmet user needs and models in PTC Creo.
- Verified GD&T equivalency to legacy devices, supporting design review and regulatory documentation.
- Automated generation of digital surgical template files using PTC Creo/MATLAB, saving 10+ hours of work.

Hardware Engineer & Computational Neuroscience Researcher

Duke University Medical Center – Brain Stimulation Engineering Lab

Jan 2024 – Present

Durham, NC

- Currently designing TMS training simulator with inductive pickup coil, front-end conditioning circuit, signal processing (amplitude and duration extraction), and display for real-time student feedback.
- Programmed patient-specific finite element analysis simulations (MATLAB/Bash) for ECT/TMS brain stimulation; designed pipelines integrating MRI image processing with mesh generation (4M+ elements) and TMS coil optimization.

Mechanical Engineering Intern

SIMS Pump Valve Company

Sep 2022 – Jun 2023

Hoboken, NJ

- Created ISO-standard CAD models/drawings using surface modeling techniques in SOLIDWORKS.
- Built standardized part library (300+ parts) for efficient modeling of pump systems in future projects.
- Performed tolerance verification on manufactured pump parts using calipers, micrometers, and 3D scanning.

PERSONAL PROJECTS | [More available at jake-golden.github.io](https://github.com/jake-golden)

MATLAB Pulse Oximeter

- Designed, built, debugged single-supply analog front-end conditioning circuit (TIA + active filtering + biased gain) converting photodiode current to PPG voltage waveform.
- Developed MATLAB DSP pipeline (filtering, amplitude extraction, R-ratio-to-SpO₂ calibration) and GUI layer (MATLAB app) with live plotting, LED control, and diagnostics console.
- Created custom Arduino-MATLAB serial protocol (fixed packets with start-byte framing and checksum validation)

PID-Controlled Self-Balancing Robot

- Built two-wheeled self-balancing robot with Arduino, IMU sensors, DC motors, custom PCB with I²C communication.
- Implemented PID controller on Arduino; iteratively tuned PID control parameters to achieve stable balancing.

SKILLS

Analog/Digital Circuit Design: LTSpice, KiCad, Siemens Xpedition, op-amp circuit design, filter design, basic CPU design, Boolean logic optimization, power electronics and power supply design (in progress)

Hardware Test & Characterization: Oscilloscopes, multimeters, SMUs, soldering, V&V test protocol development

Programming: MATLAB & Simulink, Python, C, Java, LabVIEW

Mechanical Design: SOLIDWORKS, PTC Creo, Autodesk Fusion, 3D printing (FDM), laser cutting

Applied Mathematics: Signal processing/filtering algorithms, control theory, Fourier analysis, transmission line theory, electromagnetics, hypothesis testing, inferential statistics

ACTIVITIES & INTERESTS

Duke University football student manager, classical/jazz piano, photography, college basketball enthusiast, math tutor